AI RESPONSE QUALITY EVALUATOR AGENT

Tech stack & Design Documentation

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Short description of the project :

AI Response Quality Evaluator Agent is a system designed to automatically assess the quality of AI-generated responses by evaluating their accuracy, relevance, completeness, and factual consistency while detecting hallucinations and generating performance scores.

Tech stack :

COMPONENT	TECH
Frontend	HTML/CSS/JS
Backend	FastAPI, Python 3.12
RAG	LangChain
Embeddings	Sentence-Transformers
Vector DB	ChromaDB
Grounding Datasets	TruthfulQA, SQuAD
Evaluation Frameworks	RAGAS, TruLens

Frontend: HTML/CSS/JS -- builds the interface for entering questions and viewing results

Backend: FastAPI -- connects frontend to evaluation and RAG logic via APIs

RAG: LangChain -- orchestrates retrieval, prompting, and agent coordination

Embeddings: Sentence-Transformers -- converts text into vectors for similarity search

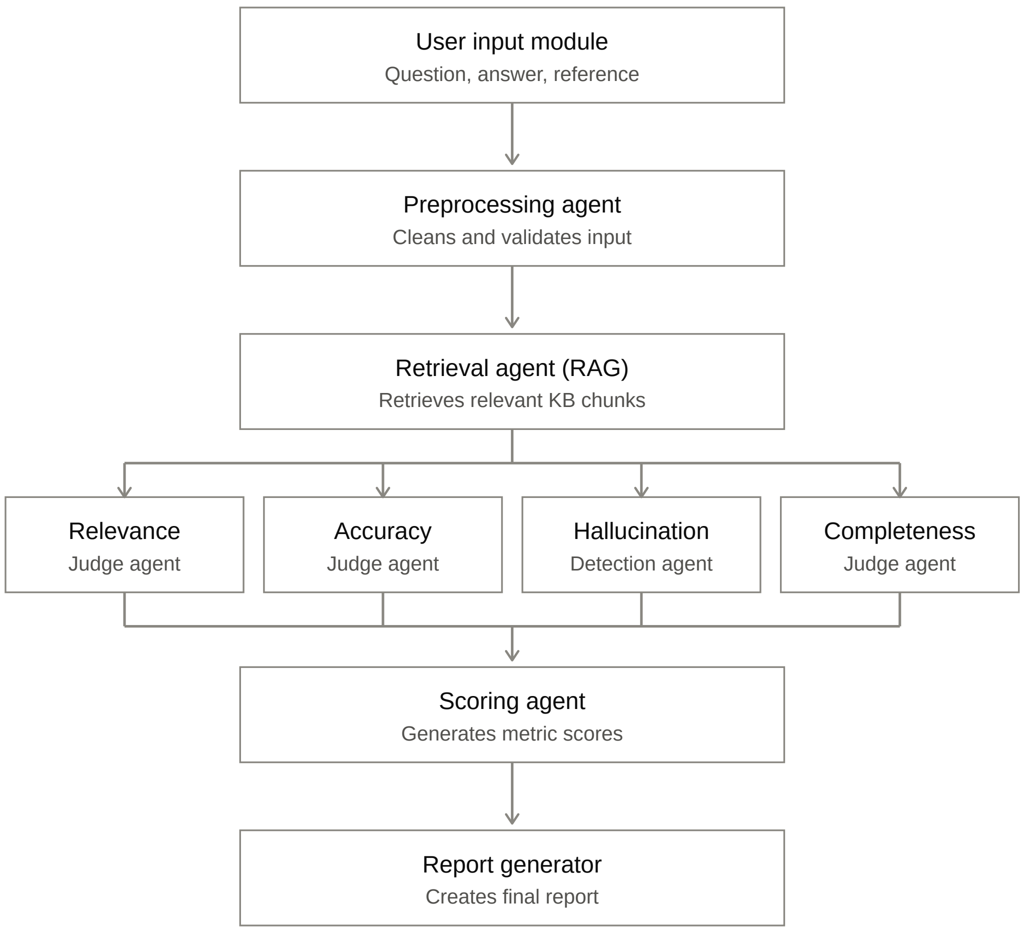
Vector Store: ChromaDB -- stores and retrieves relevant grounding context

Grounding Datasets: TruthfulQA, SQuAD -- reference data for accuracy checks

Evaluation Frameworks: RAGAS, TruLens -- score relevance, accuracy, hallucination, completeness

Design Documentation

1. High level architecture:



The pipeline starts with the **user input module**, which captures the question, the AI-generated answer, and the reference answer. This raw input then passes to the **preprocessing agent**, which cleans and validates the input to make sure it's well-formed before evaluation begins.

Next, the **retrieval agent (RAG)** retrieves relevant knowledge base chunks from ChromaDB, pulling in grounding context from TruthfulQA and SQuAD that's semantically related to the question, giving the system something factual to compare against.

From there, the retrieved context and input are passed simultaneously to four parallel evaluation agents, each judging a different quality dimension:

- The **relevance agent** checks whether the AI response actually addresses the question
- The **accuracy agent** verifies factual correctness against the retrieved context
- The **hallucination detection agent** flags any claims in the response that aren't supported by the retrieved context
- The **completeness agent** checks whether the response covers all parts of the expected answer

These four agents run independently but converge into the **scoring agent/verdict agent**, which aggregates their individual outputs into overall metric scores.

Finally, the **report generator** takes those aggregated scores and produces the final evaluation report.

2. Agent Responsibilities :

Relevance agent

- Compares the AI response against the original question
- Scores how directly and fully the response addresses what was asked
- Flags off-topic or tangential answers

Accuracy agent

- Compares the AI response against the reference answer and retrieved context
- Verifies factual correctness of claims made in the response
- Scores the degree of factual alignment

Hallucination detection agent

- Cross-checks each claim in the AI response against the retrieved context
- Identifies statements that aren't supported by any retrieved source
- Flags unsupported or fabricated content, often with a confidence score

Completeness agent

- Checks whether the response covers all parts of a multi-part question
- Compares response coverage against the expected answer's key components
- Scores how thorough the response is relative to what's expected

3. Scoring Dimensions :

Dimension	Definition	Scale
Relevance	Degree to which the response directly addresses the question asked	0–1
Accuracy	Degree to which factual claims in the response align with reference/retrieved evidence	0–1
Hallucination / Faithfulness	Proportion of response claims that are supported by retrieved evidence	0–1 (higher = more faithful)
Completeness	Whether the response covers all key aspects of the question	0–1

4. Orchestration Flow :

Step 1: Input received
The user input module captures the question, AI answer, and reference answer.

Step 2: Preprocessing (sequential)
Cleans and validates the input before anything else runs.

Step 3: Retrieval (sequential)
The retrieval agent embeds the question and fetches relevant chunks from ChromaDB (TruthfulQA/SQuAD).

Step 4: Fan-out to evaluation agents (parallel)
All four agents run simultaneously, since they're independent and only need the input + retrieved context:

- Relevance agent
- Accuracy agent
- Hallucination detection agent
- Completeness agent

Step 5: Synchronization / join
Waits for all four agents to finish. On failure or timeout, retries once or assigns a default score.

Step 6: Verdict (sequential)
Aggregates the four results into a weighted overall score.

Step 7: Report generation (sequential)
Produces the final report with reasoning and recommendations.

(Sequential just means one step happens strictly after another.)

5. Data Models :

Evaluation Request Model – Stores the user question, AI-generated response, and reference answer for evaluation.

Knowledge Base Model – Stores reference documents and text chunks used for retrieval and fact verification.

Retrieved Context Model – Stores the relevant information retrieved from the knowledge base during evaluation.

Evaluation Result Model – Stores the scores for accuracy, relevance, completeness, hallucination detection, and the final evaluation result.

Detailed data model definitions will be included in the next phase of the project.